

Cisco IOS Commands:

1. When you start the console, will see the prompt of the User EXEC mode (Router>). To see which commands are available in this mode, type “?”:
Router1> ?
2. To view and change system parameters of a Cisco router, you must enter the privileged EXEC mode, by typing:
Router1> enable
Router1#
3. To modify system wide configuration parameters, you must enter the global configuration mode. This mode is entered by typing:
Router1# configure terminal
Router1(config)#
4. To make changes to a network interface, enter the interface configuration mode, with the command:
Router1(config)# interface FastEthernet0/0
Router1(config-if)#

The name of the interface is provided as an argument. Here, the network interface that is configured is FastEthernet0/0.

5. To return from the interface configuration to the global configuration mode, or from the global configuration mode to the privileged EXEC mode, use the “exit” command:
Router1(config-if)# exit
Router1(config)# exit
Router1#

Or, to directly return to the privileged EXEC mode from any configuration mode, use the “end” command:

Router1(config-if)# end
Router1#

6. To return from the privileged EXEC mode to the user EXEC mode, type:
Router1# disable
Router1>
7. To terminate the console session from the user EXEC mode, type:
Router1> logout
Router1 con0 is now available
Press RETURN to get started.

Or type “logout” or “exit” from the privileged EXEC mode:

Router1# exit
Router1 con0 is now available
Press RETURN to get started.

In **Privileged Exec Mode** you can check your router configuration using the following commands:

interfaces
show running-config

IOS MODE: PRIVILEGED EXEC

show ip route

clear ip route *

Displays the contents of the routing table and deletes all routing table entries

show ip cache

clear ip cache

Displays the routing cache and clears the routing cache

show arp

clear arp-cache

Displays the arp cache and clears the arp cache

IOS MODE: GLOBAL CONFIGURATION

ip route-cache

Enables route caching. By default, route caching is enabled on a router.

no ip route-cache

Disables route caching

show ip ospf

Displays general information about the OSPF configuration

show ip ospf database

Displays the link state database.

show ip ospf border-routers

Displays the Area Border Router (ABR) and Autonomous System Boundary Router (ASBR).

clear ip ospf process-id process

Resets the specified OSPF process.

Adding a static host or network route via a next hop gateway:

ip route *IPAddress netmask gw_address*

ip route *NetAddress netmask gw_address*

Adding a static host or network route via a connected interface:

ip route *IPAddress netmask Iface*

ip route *NetAddress netmask Iface*

Adding a default route:

ip route 0.0.0.0 0.0.0.0 *gw_address*

Deleting static route entries:

Command same as above but use "**no ip route**" instead of "**ip route**"

IOS MODE: INTERFACE Configuration

Configuring an IP address:

Router1> enable

Router1# configure terminal

Router1(config)# no ip routing

Router1(config)# ip routing

```
Router1(config)# interface FastEthernet0/0
Router1(config-if)# ip address xxx.yyy.zzz.aaa netmask
Router1(config-if)# no shutdown
Router1(config-if)# end
```

Enabling and disabling proxy arp (enabled by default on every interface):

```
ip proxy-arp
no ip proxy-arp
```

Linux Commands

ARP Cache Related:

`arp -a`

Display the content of the ARP cache.

`arp -d IPAddress`

Deletes the entry with the IP address **IPAddress**.

`arp -s IPAddress MACAddress`

Adds a static entry to the ARP cache that is never overwritten by network events. The MAC address is entered as 6 hexadecimal bytes separated by colons.

Example: `arp -s 10.0.1.12 00:02:2D:0D:68:C1`

`ip -s -s neigh flush all`

command to clear the arp cache

Checking on the network status of the PC:

`netstat -i`

Displays a table with statistics of the currently configured network interfaces.

`netstat -rn`

Displays the kernel routing table. The `-n` option forces netstat to print the IP addresses. Without this option, netstat attempts to display the host names.

`netstat -an`

`netstat -tan`

`netstat -uan`

Displays the active network connections. The `-a` option display all active network connections, the `-ta` option displays only information on TCP connections, and the `-tu` option displays only information on UDP traffic. Omitting the `-n` option prints host names, instead of IP addresses.

`netstat -s`

Displays summary statistics for each protocol that is currently running on the host.

Configuring a PC interface:

`ifconfig`

Displays the configuration parameters of all active interfaces.

`ifconfig interface`

Displays the configuration parameters of a single interface. For example, `ifconfig eth0` displays information on interface **eth0**.

`ifconfig interface down`

Disables the interface. For example: `ifconfig eth0 down`. No traffic is sent or received on a disabled interface.

`ifconfig interface up`

Enables an interface.

`ifconfig interface IPAddr/xx`

e.g. `ifconfig eth0 10.0.1.8/24`

Assigns interface *eth0* the IP address *10.0.1.8/24* and a broadcast address of *10.0.1.255*

`ifconfig interface mtu xxx`

Assigns MTU size **xxx** bytes to **interface**

Enabling and Disabling IP Forwarding on a Linux PC:

```
sysctl net.ipv4.ip_forward
    shows current status of ipforwarding
echo 1 > '/proc/sys/net/ipv4/ip_forward'
    enables IPforwarding
echo 0 > '/proc/sys/net/ipv4/ip_forward'
    disables IPforwarding
```

Additional commands that do the same as the above commands:

```
sysctl -w net.ipv4.ip_forward=1
sysctl -w net.ipv4.ip_forward=0
```

Setting Path Discovery (MTU probing) on a Linux PC:

```
PC% echo 2 > '/proc/sys/net/ipv4/tcp_mtu_probing'
    Sets/enables probing on PC
PC% cat /proc/sys/net/ipv4/tcp_mtu_probing'
    Checks the current setting for probing. If "0" or "1" then it is not set for default probing. Has to "=2" to
    enable probing.
```

Routing commands on a PC:

```
route add -net netaddress netmask mask gw gw_address
route add -net netaddress netmask mask dev iface
    Adds a routing table entry for the network prefix identified by IP address netaddress and netmask mask.
    The next-hop is identified by IP address gw_address or by interface iface.
route add -host hostaddress gw gw_address
route add -host hostaddress dev iface
    Adds a host route entry for IP address hostaddress with the next-hop identified by IP address
gw_address or by interface iface
route add default gw gw_address
    Sets the default route to gateway with IP address gw_address
route del -net netaddress netmask mask gw gw_address
route del -host hostaddress gw gw_address
route del default gw gw_address
    Deletes an existing route from the routing table with specific arguments.
route -e
    Displays the current routing table with extended fields. The command is identical to the netstat -r
    command.
ip route flush table main
    Deletes all entries in the routing table on a PC. Please note that the local interface route(s) need to be
    added before adding any other static route entries after a flush table command. To add interface route(s)
    use the ifconfig interface down and ifconfig interface up (where e.g. interface = eth0)
ip route flush cache
    deletes all entries in the routing cache
ip route get IPAddress
ip route flush cache IPAddress
    Displays the cached route for IPAddress and flushes the cached route entry for IPAddress
```

Commonly used Linux commands:

`ping IPAddr`

Pings host with IP address **IPAddr**.

`ping IPAddr -c num`

Where **num** is the number of pings you want issued to destination **IPAddr**.

`ping IPAddr -s num`

Where **num** is the number of data bytes in the ICMP request message you want sent to destination **IPAddr**.

`traceroute IPAddr -mxxx -qyyy`

Command used to trace the route between an origin and a destination IP address **IPAddr**, where `-m` indicates the max TTL value and `-q` indicates the number of queries. E.g. `m=2`, and `q=1`.